THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 13: **Algebra Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A two-digit number exceeds the number formed by reversing its digits by 18. The difference of its digits is: 1

(a) 1 **(b)** 2 **(c)** 3 **(d)** 18

2. If $x + \frac{1}{x} = 3$, then $x^2 + \frac{1}{x^2}$ equals: 1

(a) 9 **(b)** 6 **(c)** 7 **(d)** 11

3. The sum of the dates in a 2×2 calendar block is 44. The smallest date in the block is: 1

(a) 8 **(b)** 7 **(c)** 6 **(d)** 15

4. If $a^2 - b^2 = 144$ and $a - b = 8$, then $a + b$ equals: 1

(a) 8 **(b)** 19 **(c)** 18 **(d)** 65

5. The value of $18^2 - 15 \times 21$ is: 1

(a) 9 **(b)** 18 **(c)** 0 **(d)** 6

6. The sum of three consecutive integers is 63. The largest of them is: 1

(a) 22 **(b)** 21 **(c)** 23 **(d)** 20

7. A three-digit number with digits a, b, c is reversed. If $a - c = 2$, the difference between the number and its reverse is: 1

(a) 198 **(b)** 18 **(c)** 180 **(d)** 200

8. A two-digit number exceeds the number formed by reversing its digits by 36. The difference of its digits is: 1

(a) 3 **(b)** 4 **(c)** 36 **(d)** 5

9. **Assertion (A):** For any whole number n, the number $2n$ is even and $2n + 1$ is odd. 1

Reason (R): Even numbers are multiples of 2, and odd numbers leave remainder 1 when divided by 2.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The expression $3x + 5$ has the value 17 when $x = 4$. 1
Reason (R): Terms with the same variable part are called like terms.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The sum of the digits of a two-digit number is 9 and the tens digit exceeds the ones digit by 3. Find the number. 2

12. If $x + y = 17$ and $x - y = 5$, find $x^2 - y^2$. 2

13. Prove that $(n + 3)^2 - (n - 3)^2 = 12n$ and use it to find the value of $(5 + 3)^2 - (5 - 3)^2$. 2

14. The sum of the digits of a two-digit number is 11 and the tens digit exceeds the ones digit by 5. Find the number. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The sum of a two-digit number and its reverse is 110 and the digits differ by 4. Find the number (tens digit larger). 3

16. If $x + \frac{1}{x} = 3$, find $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$. 3

17. The sum of the dates of a 2×2 block in a calendar is 36. Find the four dates and show that the block's sum is always $4\times$(smallest date)$+16$. 3

OR

The sum of a two-digit number and its reverse is 132 and the digits differ by 2. Find the number (tens digit larger). 3

18. If $x + \frac{1}{x} = 4$, find $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$. 3

OR

The sum of the dates of a 2×2 block in a calendar is 52. Find the four dates and show that the block's sum is always $4\times$(smallest date)$+16$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Aarav performs a number trick: *Think of a number n. Multiply it by 6. Add 12. Divide by 3. Subtract twice your original number. The answer is always 4.* 5
(i) Write the expression after multiplying by 6.
(ii) Write it after adding 12.
(iii) Write it after dividing by 3.
(iv) Complete the algebra to show the answer is always 4 and check with $n = 7$.

20. Prove that $(3n + 2)^2 - (3n - 2)^2 = 24n$ and use it to evaluate it for $n = 5$. 5

OR

A three-digit number has digits a, b, c with $a - c = 2$. Show that the number minus its reverse is divisible by 99 and find the difference. 5

END OF QUESTION PAPER — Total: 40 Marks